Multivariate Time Series Analysis

References:

Tsay (2010), Ch. 8; Enders (2014), Chs. 5, 6; Hamilton (1994), Chs. 10, 11, 18, 19

Detailed treatments of multivariate time series analysis are provided by:

Tsay, R. S. (2013). *Multivariate time series analysis: with R and financial applications.* John Wiley & Sons.

Lütkepohl, H. (2005). *New introduction to multiple time series analysis.* Springer Science & Business Media.

1 Weak Stationarity and Cross-correlation Matrices

1.1 Stationarity

- Multivariate time series analysis considers multiple time series simultaneously. Each variable (also referred to as component) may depend not only on its past values but also on the current and past values of other variables.
- Why consider multiple series jointly? Utilize more information; improve forecast accuracy; understand and quantify relationships between series.
- Examples of multivariate time series: consumption and income; stock prices and dividends; interest rates, money growth, and inflation.
- Notation: Vectors and matrices are useful tools and will be denoted in boldface. All vectors are column vectors.
- Consider a k-dimensional time series $\mathbf{y}_t = (y_{1t}, y_{2t}, \ldots, y_{kt})'$, for $t = 1, 2, \ldots, T$. The series $\mathbf{y}_t$ is *weakly stationary* if its first and second moments are time-invariant. That is, both

$$E(\mathbf{y}_t) = [E(y_{1t}), E(y_{2t}), \ldots, E(y_{kt})]' = \boldsymbol{\mu},$$

and

$$
\begin{aligned}
Cov\left(\mathbf{y}_t, \mathbf{y}_{t-h}\right) &= E\left[\left(\mathbf{y}_t - \boldsymbol{\mu}\right)\left(\mathbf{y}_{t-h} - \boldsymbol{\mu}\right)'\right] \\
&= \begin{bmatrix}
Cov\left(y_{1t}, y_{1,t-h}\right) & Cov\left(y_{1t}, y_{2,t-h}\right) & \cdots & Cov\left(y_{1t}, y_{k,t-h}\right) \\
Cov\left(y_{2t}, y_{1,t-h}\right) & Cov\left(y_{2t}, y_{2,t-h}\right) & \cdots & Cov\left(y_{2t}, y_{k,t-h}\right) \\
\vdots & \vdots & \cdots & \vdots \\
Cov\left(y_{kt}, y_{1,t-h}\right) & Cov\left(y_{kt}, y_{2,t-h}\right) & \cdots & Cov\left(y_{kt}, y_{k,t-h}\right)
\end{bmatrix} \\
&= \boldsymbol{\Gamma}_h,
\end{aligned} \tag{1}
$$

are time-invariant. We write $\boldsymbol{\mu} = (\mu_1, \mu_2, \ldots, \mu_k)'$ and $\boldsymbol{\Gamma}_h = [\Gamma_{ij}(h)]$ so that we can easily refer to the elements.

- When $h = 0$, $\boldsymbol{\Gamma}_0$ is the concurrent (or contemporaneous, lag zero) cross-covariance matrix of $\mathbf{y}_t$. Its ith diagonal element is the variance of y_{it}, and (i, j)th element is the covariance between y_{it} and y_{jt}. $\boldsymbol{\Gamma}_0$ is symmetric.

- In general, $\boldsymbol{\Gamma}_h$, defined in (1), is referred to as the lag-h cross-autocovariance (or cross-covariance, autocovariance) matrix. For a weakly stationary series, $\boldsymbol{\Gamma}_h$ is a function of h, but not the time index t.

- $\boldsymbol{\Gamma}_h$ is not symmetric if $h \neq 0$. For example, when $h = 1$, $\Gamma_{12}(1) = Cov\left(y_{1t}, y_{2,t-1}\right)$ and $\Gamma_{21}(1) = Cov\left(y_{2t}, y_{1,t-1}\right)$.

- The stationarity of multivariate time series implies the stationarity of individual univariate time series. The autocovariance functions of component series are the diagonal elements of $\boldsymbol{\Gamma}_h$.

1.2 Cross-correlation Matrices

- The lag-h cross-correlation matrix of $\mathbf{y}_t$ is defined as

$$
\boldsymbol{\rho}_h = [\rho_{ij}(h)] = \mathbf{D}^{-1}\boldsymbol{\Gamma}_h\mathbf{D}^{-1},
$$

where $\mathbf{D}$ is the diagonal matrix of standard deviations of individual series y_{it}, namely,

$$\begin{aligned}\mathbf{D} &= Diag\left[\text{std}\left(y_{1t}\right), \text{std}\left(y_{2t}\right), \ldots, \text{std}\left(y_{kt}\right)\right] \\ &= \begin{bmatrix} \sqrt{\Gamma_{11}\left(0\right)} & 0 & \cdots & 0 \\ 0 & \sqrt{\Gamma_{22}\left(0\right)} & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & \sqrt{\Gamma_{kk}\left(0\right)} \end{bmatrix}.\end{aligned}$$

- By definition,

$$\rho_{ij}\left(h\right) = \frac{\Gamma_{ij}\left(h\right)}{\sqrt{\Gamma_{ii}\left(0\right)\Gamma_{jj}\left(0\right)}} = \frac{Cov\left(y_{it}, y_{j,t-h}\right)}{\text{std}\left(y_{it}\right)\text{std}\left(y_{jt}\right)},$$

  which is the correlation coefficient between y_{it} and $y_{j,t-h}$. Also note that the ith diagonal element of $\boldsymbol{\rho}_h$, $\rho_{ii}\left(h\right)$, is the lag-h autocorrelation coefficient of y_{it}.

- Key properties of cross-covariances and cross-correlations when $h > 0$:
  - $\boldsymbol{\Gamma}_h$ and $\boldsymbol{\rho}_h$ are in general not symmetric.
  - Under stationarity, $\boldsymbol{\Gamma}_h = \boldsymbol{\Gamma}'_{-h}$ and $\boldsymbol{\rho}_h = \boldsymbol{\rho}'_{-h}$. To see this, for $1 \leq i, j \leq k$, we have

$$Cov\left(y_{it}, y_{j,t-h}\right) = Cov\left(y_{j,t-h}, y_{it}\right) = Cov\left(y_{jt}, y_{i,t+h}\right) = Cov\left(y_{jt}, y_{i,t-(-h)}\right),$$

    and hence $\Gamma_{ij}\left(h\right) = \Gamma_{ji}\left(-h\right)$. The result on $\boldsymbol{\rho}_h$ immediately follows, and accordingly in practice we only need to consider the cross-correlation matrices $\boldsymbol{\rho}_h$ for $h \geq 0$.

- The *linear* relationship between two series $\{y_{it}\}$ and $\{y_{jt}\}$ $(i \neq j)$ can be summarized as follows (Tsay, 2010, p. 392):
  - y_{it} and y_{jt} have *no linear relationship* if $\rho_{ij}\left(h\right) = \rho_{ji}\left(h\right) = 0$ for all $h \geq 0$.
  - y_{it} and y_{jt} are *concurrently correlated* if $\rho_{ij}\left(0\right) \neq 0$.
  - y_{it} and y_{jt} have *no lead-lag relationship* (aka *uncoupled*) if $\rho_{ij}\left(h\right) = 0$ and $\rho_{ji}\left(h\right) = 0$ for all $h > 0$.

- There is a *unidirectional relationship* from y_{it} to y_{jt} if $\rho_{ij}(h) = 0$ for all $h > 0$, but $\rho_{ji}(l) \neq 0$ for some $l > 0$. That is, y_{it} does not depend on any past value of y_{jt}, but y_{jt} depends on some past values of y_{it}.
- There is a *feedback relationship* between y_{it} and y_{jt} if $\rho_{ij}(h) \neq 0$ for some $h > 0$, and $\rho_{ji}(l) \neq 0$ for some $l > 0$.

- $\mathbf{\Gamma}_h$ can be estimated by the sample cross-covariance matrix, which is given by

$$\hat{\mathbf{\Gamma}}_h = \frac{1}{T}\sum_{t=h+1}^{T}(\mathbf{y}_t - \bar{\mathbf{y}})(\mathbf{y}_{t-h} - \bar{\mathbf{y}})', \quad \text{for } h \geq 0,$$

where $\bar{\mathbf{y}} = T^{-1}\sum_{t=1}^{T}\mathbf{y}_t$ is the vector of sample means, and $\boldsymbol{\rho}_h$ can be estimated by

$$\hat{\boldsymbol{\rho}}_h = \hat{\mathbf{D}}^{-1}\hat{\mathbf{\Gamma}}_h\hat{\mathbf{D}}^{-1}, \quad \text{for } h \geq 0,$$

where $\hat{\mathbf{D}}$ is the diagonal matrix of sample standard deviations of the component series.

1.3 Multivariate Portmanteau Tests

- The univariate Ljung-Box statistic can be generalized to the multivariate case (Hosking, 1980, 1981).

- For a k-dimensional multivariate series $\mathbf{y}_t$, the statistic for testing $H_0 : \boldsymbol{\rho}_1 = \boldsymbol{\rho}_2 = \ldots = \boldsymbol{\rho}_m = \mathbf{0}$ against $H_1 : \boldsymbol{\rho}_i \neq \mathbf{0}$, for some $i \in \{1, 2, \ldots, m\}$, is

$$Q_k(m) = T^2\sum_{h=1}^{m}\frac{1}{T-h}tr\left(\hat{\mathbf{\Gamma}}_h'\hat{\mathbf{\Gamma}}_0^{-1}\hat{\mathbf{\Gamma}}_h\hat{\mathbf{\Gamma}}_0^{-1}\right),$$

where T is the sample size and $tr(\mathbf{A})$ denotes the trace of a matrix $\mathbf{A}$.

- Under H_0 and some regularity conditions, $Q_k(m) \sim \chi^2(k^2 m)$.

- If H_0 is rejected, we may build a multivariate model for $\mathbf{y}_t$ to study the lead-lag relationships between the component series.

2 Vector Autoregressive Models

2.1 VAR(1) Model

2.1.1 Model Specification

- A $k \times 1$ vector of random variables, $\mathbf{y}_t$, is said to follow a *vector autoregressive model of order 1*, or VAR(1) for short, if

$$\mathbf{y}_t = \boldsymbol{\phi}_0 + \boldsymbol{\Phi}\mathbf{y}_{t-1} + \boldsymbol{\varepsilon}_t, \quad t = 1, 2, \ldots, \tag{2}$$

  where $\boldsymbol{\phi}_0$ is a $k \times 1$ vector, $\boldsymbol{\Phi}$ is a $k \times k$ matrix, and $\boldsymbol{\varepsilon}_t$ is a $k \times 1$ vector of disturbances that satisfy the following assumptions.

  - $E(\boldsymbol{\varepsilon}_t) = \mathbf{0}$, for all t
  - $E(\boldsymbol{\varepsilon}_t \boldsymbol{\varepsilon}_t') = \boldsymbol{\Sigma}$, where $\boldsymbol{\Sigma}$ is a $k \times k$ positive definite matrix
  - $E(\boldsymbol{\varepsilon}_t \boldsymbol{\varepsilon}_{t'}') = \mathbf{0}$, for all $t \neq t'$

  We write $\boldsymbol{\varepsilon}_t \sim WN(\mathbf{0}, \boldsymbol{\Sigma})$ as a shorthand as before.

- For example, when $k = 2$, the VAR(1) model is

$$\begin{pmatrix} y_{1t} \\ y_{2t} \end{pmatrix} = \begin{pmatrix} \phi_{10} \\ \phi_{20} \end{pmatrix} + \begin{pmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{pmatrix} \begin{pmatrix} y_{1,t-1} \\ y_{2,t-1} \end{pmatrix} + \begin{pmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{pmatrix},$$

  or equation by equation

$$y_{1t} = \phi_{10} + \phi_{11} y_{1,t-1} + \phi_{12} y_{2,t-1} + \varepsilon_{1t}, \tag{3}$$

$$y_{2t} = \phi_{20} + \phi_{21} y_{1,t-1} + \phi_{22} y_{2,t-1} + \varepsilon_{2t}, \tag{4}$$

  where $\boldsymbol{\phi}_0 = (\phi_{10}, \phi_{20})'$, $\boldsymbol{\varepsilon}_t = (\varepsilon_{1t}, \varepsilon_{2t})'$, ϕ_{ij} is the (i,j)th element of $\boldsymbol{\Phi}$, for $i, j = 1, 2$, and

$$\boldsymbol{\Sigma} = E(\boldsymbol{\varepsilon}_t \boldsymbol{\varepsilon}_t') = \begin{pmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{21} & \sigma_2^2 \end{pmatrix},$$

where $\sigma_{12} = \sigma_{21}$.

- $\mathbf{\Phi}$ measures the dynamic dependence of $\mathbf{y}_t$. Note that ϕ_{11} and ϕ_{12} represent the (linear) dependence of y_{1t} on its own past, $y_{1,t-1}$, and on the past of the other variable, $y_{2,t-1}$, respectively. To put it another way, ϕ_{12} is the conditional effect of $y_{2,t-1}$ on y_{1t} given $y_{1,t-1}$.

- *Exercise*: Consider Equations (3) and (4) jointly.

  - (a) Under what conditions will there be a *unidirectional* relationship from y_{1t} to y_{2t}?
  - (b) Under what conditions will y_{1t} and y_{2t} be *uncoupled*?
  - (c) Under what conditions will there be a *feedback* relationship between y_{1t} and y_{2t}?

- The *concurrent* relationship between y_{1t} and y_{2t} is measured by the off-diagonal elements of $\mathbf{\Sigma}$. If $\sigma_{12} = 0$, then the two-component series are not concurrently correlated.

- The VAR model is a *reduced-form* model since it does not explicitly show the concurrent dependence between component series. If desired, an explicit representation (*structural equation*) involving the concurrent dependence can be deduced from the reduced-form model by a simple linear transformation (see, Tsay, 2010, Section 8.2.1).

- The reduced-form model avoids endogeneity by using lagged values as explanatory variables. It can be easily estimated and used in forecasting. In contrast, concurrent models have simultaneity bias issues in estimation and cannot be used in forecasting.

- The VAR model is a special case of *seemingly unrelated regression* (SUR) models with lagged variables (and deterministic terms) as regressors.

2.1.2 Stationarity Condition

- Suppose that the VAR(1) model given by (2) is weakly stationary. Taking expectation of both sides of the equation yields

$$E(\mathbf{y}_t) = \boldsymbol{\phi}_0 + \mathbf{\Phi} E(\mathbf{y}_{t-1}) + E(\boldsymbol{\varepsilon}_t).$$

Using $E(\boldsymbol{\varepsilon}_t) = \mathbf{0}$ and $E(\mathbf{y}_t) = E(\mathbf{y}_{t-1})$, we have

$$\boldsymbol{\mu} = E(\mathbf{y}_t) = (\mathbf{I}_k - \boldsymbol{\Phi})^{-1} \boldsymbol{\phi}_0, \tag{5}$$

assuming that $\mathbf{I}_k - \boldsymbol{\Phi}$ is nonsingular, where $\mathbf{I}_k$ is the $k \times k$ identity matrix.

- (5) implies $\boldsymbol{\phi}_0 = (\mathbf{I}_k - \boldsymbol{\Phi})\boldsymbol{\mu}$. Using this result in Model (2) gives

$$\mathbf{y}_t - \boldsymbol{\mu} = \boldsymbol{\Phi}(\mathbf{y}_{t-1} - \boldsymbol{\mu}) + \boldsymbol{\varepsilon}_t.$$

Let $\tilde{\mathbf{y}}_t = \mathbf{y}_t - \boldsymbol{\mu}$. Then the VAR(1) model can be written as

$$\tilde{\mathbf{y}}_t = \boldsymbol{\Phi}\tilde{\mathbf{y}}_{t-1} + \boldsymbol{\varepsilon}_t. \tag{6}$$

- By repeated substitutions, we obtain the MA(∞) representation of (6):

$$\tilde{\mathbf{y}}_t = \boldsymbol{\varepsilon}_t + \boldsymbol{\Phi}\boldsymbol{\varepsilon}_{t-1} + \boldsymbol{\Phi}^2\boldsymbol{\varepsilon}_{t-2} + \boldsymbol{\Phi}^3\boldsymbol{\varepsilon}_{t-3} + \ldots = \sum_{h=0}^{\infty} \boldsymbol{\Phi}^h \boldsymbol{\varepsilon}_{t-h}, \tag{7}$$

where $\boldsymbol{\Phi}^0 \equiv \mathbf{I}_k$.

- Properties of VAR(1) can be seen from (7):
  - $Cov(\boldsymbol{\varepsilon}_t, \mathbf{y}_{t-h}) = \mathbf{0}$, for all $h > 0$, since $\boldsymbol{\varepsilon}_t$ is serially uncorrelated. Therefore, $\boldsymbol{\varepsilon}_t$ is referred to as the *shock* or *innovation* of the series at time t.
  - $Cov(\boldsymbol{\varepsilon}_t, \mathbf{y}_t) = \boldsymbol{\Sigma}$. This can be seen by post-multiplying both sides of (7) by $\boldsymbol{\varepsilon}_t'$ and taking expectations, namely,

$$E(\tilde{\mathbf{y}}_t\boldsymbol{\varepsilon}_t') = E(\boldsymbol{\varepsilon}_t\boldsymbol{\varepsilon}_t') + \boldsymbol{\Phi}E(\boldsymbol{\varepsilon}_t\boldsymbol{\varepsilon}_{t-1}') + \boldsymbol{\Phi}^2E(\boldsymbol{\varepsilon}_t\boldsymbol{\varepsilon}_{t-2}') + \ldots.$$

  - $\mathbf{y}_t$ depends on the past shock $\boldsymbol{\varepsilon}_{t-h}$ by the coefficient matrix $\boldsymbol{\Phi}^h$. For such dependence to decay, $\boldsymbol{\Phi}^h$ must converge to zero as $h \to \infty$. This will be guaranteed if all the k eigenvalues of $\boldsymbol{\Phi}$ are less than one in modulus.

- $\mathbf{y}_t$ is said to be stable if all eigenvalues of $\mathbf{\Phi}$ are less than one in modulus (assuming $\mathbf{\Sigma}$ exists). Recall that the eigenvalues of $\mathbf{\Phi}$, denoted by λ, are defined by

$$|\mathbf{\Phi} - \lambda \mathbf{I}_k| = 0. \tag{8}$$

  The *stability condition* can be rephrased as all the roots of the determinant equation (8) lying inside the unit circle ($|\lambda| < 1$).

- A stable VAR process is stationarity, but an unstable process is not necessarily nonstationary. The stability condition is often called the *stationarity condition* in the time series literature.

- Since

$$|\mathbf{\Phi} - \lambda \mathbf{I}_k| = \lambda^k \left| \mathbf{I}_k - \mathbf{\Phi} \frac{1}{\lambda} \right|,$$

  the stationarity condition is equivalent to all the roots of $|\mathbf{I}_k - \mathbf{\Phi} z| = 0$ lying outside the unit circle, where $z = 1/\lambda$.

- *Example*: Consider the bivariate VAR(1) model

$$\mathbf{y}_t = \boldsymbol{\phi}_0 + \mathbf{\Phi} \mathbf{y}_{t-1} + \boldsymbol{\varepsilon}_t,$$

  or

$$\begin{pmatrix} y_{1t} \\ y_{2t} \end{pmatrix} = \begin{pmatrix} \phi_{10} \\ \phi_{20} \end{pmatrix} + \begin{pmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{pmatrix} \begin{pmatrix} y_{1,t-1} \\ y_{2,t-1} \end{pmatrix} + \begin{pmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{pmatrix}.$$

  Then $\mathbf{y}_t$ is a weakly stationary VAR(1) process if the roots of the following equation lie inside the unit circle ($|\lambda| < 1$):

$$\begin{aligned} |\mathbf{\Phi} - \lambda \mathbf{I}_2| &= det \left[\begin{pmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right] = det \begin{pmatrix} \phi_{11} - \lambda & \phi_{12} \\ \phi_{21} & \phi_{22} - \lambda \end{pmatrix} \\ &= (\phi_{11} - \lambda)(\phi_{22} - \lambda) - \phi_{12}\phi_{21} = 0. \end{aligned}$$

  This stationarity condition involves cross-terms ϕ_{12} and ϕ_{21}. In contrast, if $\phi_{12} = \phi_{21} = 0$,

we have $\lambda_1 = \phi_{11}$ and $\lambda_2 = \phi_{22}$, and then the bivariate stationarity condition reduces to the univariate stationarity conditions for each equation.

2.1.3 Cross Covariances and Correlations of VAR(1)

- Using the MA representation, (7), the (concurrent) covariance matrix of $\mathbf{y}_t$ is given by

$$Cov\left(\mathbf{y}_t\right) = \mathbf{\Gamma}_0 = \mathbf{\Sigma} + \mathbf{\Phi}\mathbf{\Sigma}\mathbf{\Phi}' + \mathbf{\Phi}^2\mathbf{\Sigma}\left(\mathbf{\Phi}^2\right)' + \ldots = \sum_{i=0}^{\infty} \mathbf{\Phi}^i \mathbf{\Sigma}\left(\mathbf{\Phi}^i\right)' < \infty,$$

  where $\mathbf{\Phi}^0 \equiv \mathbf{I}_k$.

- Post-multiplying both sides of (6), $\tilde{\mathbf{y}}_t = \mathbf{\Phi}\tilde{\mathbf{y}}_{t-1} + \boldsymbol{\varepsilon}_t$, by $\tilde{\mathbf{y}}'_{t-h}$ and taking expectation yields

$$E\left(\tilde{\mathbf{y}}_t\tilde{\mathbf{y}}'_{t-h}\right) = \mathbf{\Phi}E\left(\tilde{\mathbf{y}}_{t-1}\tilde{\mathbf{y}}'_{t-h}\right) + E\left(\boldsymbol{\varepsilon}_t\tilde{\mathbf{y}}'_{t-h}\right).$$

  Recall that $Cov\left(\boldsymbol{\varepsilon}_t, \mathbf{y}_{t-h}\right) = E\left(\boldsymbol{\varepsilon}_t\tilde{\mathbf{y}}'_{t-h}\right) = \mathbf{0}$, for all $h > 0$. Then,

$$\mathbf{\Gamma}_h = \mathbf{\Phi}\mathbf{\Gamma}_{h-1}, \quad h > 0, \tag{9}$$

  where $\mathbf{\Gamma}_h$ is the lag-h cross-covariance matrix of $\mathbf{y}_t$. By repeated substitutions,

$$\mathbf{\Gamma}_h = \mathbf{\Phi}^h\mathbf{\Gamma}_0, \quad h > 0.$$

- Pre- and post-multiplying (9) by $\mathbf{D}^{-1}$, we can show that $\boldsymbol{\rho}_h = \boldsymbol{\Upsilon}\boldsymbol{\rho}_{h-1}$, for $h > 0$, where $\boldsymbol{\Upsilon} = \mathbf{D}^{-1}\mathbf{\Phi}\mathbf{D}$, and hence $\boldsymbol{\rho}_h = \boldsymbol{\Upsilon}^h\boldsymbol{\rho}_0$, for $h > 0$.

2.2 VAR(p) Models

- $\mathbf{y}_t$ follows a VAR(p) model if

$$\mathbf{y}_t = \boldsymbol{\phi}_0 + \mathbf{\Phi}_1\mathbf{y}_{t-1} + \mathbf{\Phi}_2\mathbf{y}_{t-2} + \ldots + \mathbf{\Phi}_p\mathbf{y}_{t-p} + \boldsymbol{\varepsilon}_t, \tag{10}$$

  where $\boldsymbol{\phi}_0$ is a $k \times 1$ vector, $\mathbf{\Phi}_j$ are $k \times k$ matrices (with $\mathbf{\Phi}_p \neq \mathbf{0}$), and $\boldsymbol{\varepsilon}_t \sim WN\left(\mathbf{0}, \mathbf{\Sigma}\right)$.

- Model (10) can be represented using the lag operators as $\mathbf{\Phi}(L)\,\mathbf{y}_t = \boldsymbol{\phi}_0 + \boldsymbol{\varepsilon}_t$, where $\mathbf{\Phi}(L) = \mathbf{I}_k - \mathbf{\Phi}_1 L - \mathbf{\Phi}_2 L^2 - \ldots - \mathbf{\Phi}_p L^p$ is a matrix polynomial.

- *Companion form* of the VAR(p) model: Model (10) can be rewritten as

$$\begin{pmatrix} \mathbf{y}_t \\ \mathbf{y}_{t-1} \\ \vdots \\ \mathbf{y}_{t-p+2} \\ \mathbf{y}_{t-p+1} \end{pmatrix} = \begin{pmatrix} \boldsymbol{\phi}_0 \\ \mathbf{0} \\ \vdots \\ \mathbf{0} \\ \mathbf{0} \end{pmatrix} + \begin{pmatrix} \mathbf{\Phi}_1 & \mathbf{\Phi}_2 & \ldots & \mathbf{\Phi}_{p-1} & \mathbf{\Phi}_p \\ \mathbf{I}_k & \mathbf{0} & \ldots & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_k & \ldots & \mathbf{0} & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \mathbf{0} & \mathbf{0} & \ldots & \mathbf{I}_k & \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{y}_{t-1} \\ \mathbf{y}_{t-2} \\ \vdots \\ \mathbf{y}_{t-p+1} \\ \mathbf{y}_{t-p} \end{pmatrix} + \begin{pmatrix} \boldsymbol{\varepsilon}_t \\ \mathbf{0} \\ \vdots \\ \mathbf{0} \\ \mathbf{0} \end{pmatrix},$$

  which is a VAR(1) model for $\mathbf{Y}_t$:

$$\mathbf{Y}_t = \boldsymbol{\alpha} + \mathbf{\Phi}\mathbf{Y}_{t-1} + \mathbf{U}_t,$$

  where $\mathbf{Y}_t = \left(\mathbf{y}_t', \mathbf{y}_{t-1}', \ldots, \mathbf{y}_{t-p+1}'\right)'$, $\boldsymbol{\alpha} = (\boldsymbol{\phi}_0', \mathbf{0}, \ldots, \mathbf{0})'$, and $\mathbf{U}_t = (\boldsymbol{\varepsilon}_t', \mathbf{0}, \ldots, \mathbf{0})'$ are $kp \times 1$ vectors, and

$$\underset{kp\times kp}{\mathbf{\Phi}} = \begin{pmatrix} \mathbf{\Phi}_1 & \mathbf{\Phi}_2 & \ldots & \mathbf{\Phi}_{p-1} & \mathbf{\Phi}_p \\ \mathbf{I}_k & \mathbf{0} & \ldots & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_k & \ldots & \mathbf{0} & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \mathbf{0} & \mathbf{0} & \ldots & \mathbf{I}_k & \mathbf{0} \end{pmatrix}.$$

  The properties of the VAR(p) model can be derived from the results of a VAR(1) model.

- The VAR(p) model given by (10) is weakly stationary if all the eigenvalues of $\mathbf{\Phi}$ are less than one in modulus. Since the eigenvalues of $\mathbf{\Phi}$ satisfy[1]

$$|\mathbf{\Phi} - \lambda \mathbf{I}_{kp}| = \left|\mathbf{I}_k \lambda^p - \mathbf{\Phi}_1 \lambda^{p-1} - \mathbf{\Phi}_2 \lambda^{p-2} - \ldots - \mathbf{\Phi}_p\right| = 0.$$

  Hence, the sufficient condition for weak stationarity for the VAR(p) model is that all roots of the above determinant equation lie inside the unit circle ($|\lambda| < 1$), or equivalently, all

[1] See Hamilton (1994) Appendix 10.A for a proof.

roots of the equation

$$|\mathbf{I}_{kp} - \mathbf{\Phi} z| = \left|\mathbf{I}_k - \mathbf{\Phi}_1 z - \mathbf{\Phi}_2 z^2 - \ldots - \mathbf{\Phi}_p z^p\right| = 0$$

lie outside the unit circle ($|z| > 1$).

- If $\mathbf{y}_t$ is weakly stationary, then
  - $\boldsymbol{\mu} = E(\mathbf{y}_t) = (\mathbf{I}_k - \mathbf{\Phi}_1 - \mathbf{\Phi}_2 - \ldots - \mathbf{\Phi}_p)^{-1} \boldsymbol{\phi}_0 = [\mathbf{\Phi}(1)]^{-1} \boldsymbol{\phi}_0$
  - $Cov(\mathbf{y}_t, \boldsymbol{\varepsilon}_t) = \mathbf{\Sigma}$, $Cov(\mathbf{y}_{t-h}, \boldsymbol{\varepsilon}_t) = \mathbf{0}$, for $h > 0$
  - $\mathbf{\Gamma}_h = \mathbf{\Phi}_1 \mathbf{\Gamma}_{h-1} + \mathbf{\Phi}_2 \mathbf{\Gamma}_{h-2} + \ldots + \mathbf{\Phi}_p \mathbf{\Gamma}_{h-p}$, for $h > 0$. This property is known as the moment equations of a VAR(p) model, which is a multivariate version of the Yule-Walker equations of a univariate AR(p) model. Of course, it can also be expressed in terms of the cross-correlation matrices.

2.3 The Moving Average Representation of a VAR Process

- Similar to the univariate case, any weakly stationary multivariate time series can be written as a linear function of the past and present innovations.
- **Multivariate Wold Representation**: Any $k \times 1$ weakly stationary multivariate time series $\mathbf{y}_t$ has a moving average (aka Wold, or linear process) representation of the form

$$\mathbf{y}_t = \boldsymbol{\mu} + \boldsymbol{\varepsilon}_t + \mathbf{\Psi}_1 \boldsymbol{\varepsilon}_{t-1} + \mathbf{\Psi}_2 \boldsymbol{\varepsilon}_{t-2} + \ldots = \boldsymbol{\mu} + \sum_{i=0}^{\infty} \mathbf{\Psi}_i \boldsymbol{\varepsilon}_{t-i}, \tag{11}$$

where $\boldsymbol{\varepsilon}_t \sim WN(\mathbf{0}, \mathbf{\Sigma})$, $\mathbf{\Psi}_0 \equiv \mathbf{I}_k$, and $\mathbf{\Psi}_i$ are $k \times k$ matrices.

- $\mathbf{\Psi}_i$ is the effect of past innovation $\boldsymbol{\varepsilon}_{t-i}$ on $\mathbf{y}_t$, or equivalently, the effect of $\boldsymbol{\varepsilon}_t$ on $\mathbf{y}_{t+i}$. Accordingly, $\mathbf{\Psi}_i$ is often referred to as the *impulse response function* of $\mathbf{y}_t$.
- For the stationary VAR(p) model, the MA coefficients $\mathbf{\Psi}_i$ can be determined from the VAR coefficients $\mathbf{\Phi}_i$ by solving

$$\left(\mathbf{I}_k - \mathbf{\Phi}_1 L - \mathbf{\Phi}_2 L^2 - \ldots - \mathbf{\Phi}_p L^p\right)\left(\mathbf{I}_k + \mathbf{\Psi}_1 L + \mathbf{\Psi}_2 L^2 + \ldots\right) = \mathbf{I}_k, \tag{12}$$

where, as before, L is the lag operator defined such that $L\mathbf{y}_t = \mathbf{y}_{t-1}$. Equation (12) can be written more compactly as

$$\mathbf{\Phi}(L)\,\mathbf{\Psi}(L) = \mathbf{I}_k, \quad \text{or } \mathbf{\Psi}(L) = \mathbf{\Phi}(L)^{-1},$$

where $\mathbf{\Phi}(L) = \mathbf{I}_k - \mathbf{\Phi}_1 L - \ldots - \mathbf{\Phi}_p L^p$, and $\mathbf{\Psi}(L) = \sum_{i=0}^{\infty} \mathbf{\Psi}_i L^i$ with $\mathbf{\Psi}_0 = \mathbf{I}_k$. Equating the coefficients on both sides of (12) gives

$$\begin{aligned}
\mathbf{\Psi}_1 &= \mathbf{\Phi}_1, \\
\mathbf{\Psi}_2 &= \mathbf{\Phi}_1\mathbf{\Psi}_1 + \mathbf{\Phi}_2, \\
&\vdots \\
\mathbf{\Psi}_i &= \mathbf{\Phi}_1\mathbf{\Psi}_{i-1} + \ldots + \mathbf{\Phi}_p\mathbf{\Psi}_{i-p}.
\end{aligned}$$

In summary, $\mathbf{\Psi}_i$ can be computed recursively using

$$\mathbf{\Psi}_i = \sum_{j=1}^{i} \mathbf{\Phi}_j\mathbf{\Psi}_{i-j}, \quad i = 1, 2, \ldots, \text{ with } \mathbf{\Psi}_0 = \mathbf{I}_k.$$

- The mean vector can be obtained as $\boldsymbol{\mu} = \mathbf{\Psi}(1)\,\boldsymbol{\phi}_0 = \mathbf{\Phi}(1)^{-1}\,\boldsymbol{\phi}_0 = (\mathbf{I}_k - \mathbf{\Phi}_1 - \ldots - \mathbf{\Phi}_p)^{-1}\,\boldsymbol{\phi}_0$.

2.4 Estimation

- Assume that the VAR(p) process given by (10) is weakly stationary, and there are no restrictions on the parameters. Suppose we have observations on $\mathbf{y}_t$ for $(T + p)$ periods. As in the univariate case, the simplest approach is to condition on the first p observations, denoted by $\mathbf{y}_{-p+1}, \mathbf{y}_{-p+2}, \ldots, \mathbf{y}_0$, and to base estimation on the last T observations, $\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_T$.

- It is convenient to rewrite the model more compactly as

$$\mathbf{y}_t = \boldsymbol{\phi}_0 + \mathbf{\Phi}_1\mathbf{y}_{t-1} + \mathbf{\Phi}_2\mathbf{y}_{t-2} + \ldots + \mathbf{\Phi}_p\mathbf{y}_{t-p} + \boldsymbol{\varepsilon}_t \equiv \mathbf{\Pi}'\mathbf{x}_t + \boldsymbol{\varepsilon}_t,$$

where

$$\underset{k\times(kp+1)}{\mathbf{\Pi}'} = (\boldsymbol{\phi}_0, \mathbf{\Phi}_1, \mathbf{\Phi}_2, \ldots, \mathbf{\Phi}_p), \qquad \underset{(kp+1)\times 1}{\mathbf{x}_t} = \begin{pmatrix} 1 \\ \mathbf{y}_{t-1} \\ \mathbf{y}_{t-2} \\ \vdots \\ \mathbf{y}_{t-p} \end{pmatrix}.$$

- If $\boldsymbol{\varepsilon}_t \sim IIDN(\mathbf{0}, \mathbf{\Sigma})$, then

$$\mathbf{y}_t | \mathbf{y}_{t-1}, \mathbf{y}_{t-2}, \ldots, \mathbf{y}_{t-p} \sim N(\mathbf{\Pi}'\mathbf{x}_t, \mathbf{\Sigma}).$$

- The objective is to maximize the log likelihood for the sample conditional on the initial values $\mathbf{y}_0, \mathbf{y}_{-1}, \ldots, \mathbf{y}_{-p+1}$:

$$\begin{aligned} \log L(\boldsymbol{\theta}) &= \sum_{t=1}^{T} \log f(\mathbf{y}_t | \mathbf{y}_{t-1}, \mathbf{y}_{t-2}, \ldots, \mathbf{y}_{t-p}; \boldsymbol{\theta}) \\ &= -\frac{Tk}{n}\log(2\pi) + \frac{T}{2}\log\left|\mathbf{\Sigma}^{-1}\right| - \frac{1}{2}\sum_{t=1}^{T}\left[(\mathbf{y}_t - \mathbf{\Pi}'\mathbf{x}_t)'\mathbf{\Sigma}^{-1}(\mathbf{y}_t - \mathbf{\Pi}'\mathbf{x}_t)\right], \end{aligned}$$

  where $\boldsymbol{\theta}$ is a vector that contains all the elements of $\mathbf{\Pi}$ and $\mathbf{\Sigma}$. Then the conditional maximum likelihood estimator (CLME) of $\mathbf{\Pi}'$ can be computed by

$$\underset{k\times(kp+1)}{\hat{\mathbf{\Pi}}'} = \left(\sum_{t=1}^{T}\mathbf{y}_t\mathbf{x}_t'\right)\left(\sum_{t=1}^{T}\mathbf{x}_t\mathbf{x}_t'\right)^{-1},$$

  which is the same as the system OLS estimator. The jth row of $\mathbf{\Pi}'$ is

$$\underset{1\times(kp+1)}{\hat{\boldsymbol{\pi}}'} = \left(\sum_{t=1}^{T}y_{jt}\mathbf{x}_t'\right)\left(\sum_{t=1}^{T}\mathbf{x}_t\mathbf{x}_t'\right)^{-1},$$

  which is just the OLS estimator of the coefficient vector from regressing y_{jt} on $\mathbf{x}_t = \left(1, \mathbf{y}_{t-1}', \mathbf{y}_{t-2}', \ldots, \mathbf{y}_{t-p}'\right)'$.

- In sum, the CLME is asymptotically equivalent to the system OLS, which is the same as OLS equation-by-equation since the VAR(p) model is a seemingly unrelated regression

(SUR) model.

- The OLS estimates are consistent and asymptotically efficient under regularity conditions. Even though the errors are correlated across equations, the system feasible generalized least squares (FGLS) estimation (aka SUR) does not improve efficiency over OLS equation-by-equation because all equations have identical regressors.

- The CMLE of $\mathbf{\Sigma}$ is

$$\hat{\mathbf{\Sigma}} = \frac{1}{T}\sum_{t=1}^{T} \hat{\boldsymbol{\varepsilon}}_t \hat{\boldsymbol{\varepsilon}}_t', \quad \text{where } \hat{\boldsymbol{\varepsilon}}_t = \mathbf{y}_t - \hat{\mathbf{\Pi}}' \mathbf{x}_t,$$

  which is different from the least squares estimate:

$$\hat{\mathbf{\Sigma}} = \frac{1}{T-p-1}\sum_{t=1}^{T} \hat{\boldsymbol{\varepsilon}}_t \hat{\boldsymbol{\varepsilon}}_t'.$$

- The CMLE consistently estimates $\mathbf{\Pi}$ and $\mathbf{\Sigma}$ even if the true innovations are non-Gaussian, provided that the mean equations are correctly specified.

- The standard errors of $\hat{\mathbf{\Pi}}$ can be obtained using the usual OLS formulas. For a detailed representation of the asymptotic distribution of $\hat{\mathbf{\Pi}}$ and $\hat{\mathbf{\Sigma}}$, see Hamilton (1994, pp. 298–302) .

2.5 Forecasting

- The steps for constructing a stationary VAR(p) model are similar to those for the univariate case.

  - Order selection: The order p can be selected by the information criteria (such as AIC and BIC) or sequential likelihood ratio tests.
  - Estimation: OLS/ML. They are asymptotically equivalent and normally distributed under some regularity conditions.
  - Residual diagnostics: $Q_k(m) \sim \chi^2\left(k^2(m-p)\right)$, where m is the maximum lag order included in the test, and p is the order of the VAR. (One can also check conditional heteroskedasticity and normality of the residual series if needed.)

– Forecasting: The fitted model can be used for forecasting if it is adequate.

- We can apply the same techniques as those in the univariate case to produce forecasts and variances of the associated forecast errors.

- Assuming the quadratic loss function, the optimal forecast (aka the minimum MSE forecast) for $\mathbf{y}_{t+h}$ at forecast origin t is $\hat{\mathbf{y}}_{t+h|t} = E(\mathbf{y}_{t+h}|\Omega_t)$.

- For the VAR(p) model given by (10), the h-step-ahead forecasts can be obtained by iterating on

$$\hat{\mathbf{y}}_{t+h|t} = \boldsymbol{\phi}_0 + \sum_{i=1}^{p} \boldsymbol{\Phi}_i \hat{\mathbf{y}}_{t+h-i|t}, \text{ for } h = 1, 2, \ldots,$$

where the initial values are $\hat{\mathbf{y}}_{t+h-i|t} = \mathbf{y}_{t-i}$, for $h - i \leq 0$.

- If $\mathbf{y}_t$ is weakly stationary, the h-step-ahead forecast converges to its mean vector, $\boldsymbol{\mu}$, as $h \to \infty$, and the covariance matrix of its forecast error converges to the covariance matrix of $\mathbf{y}_t$.

2.6 Granger Causality

- An interesting notion that arises from the multivariate time series analysis is Granger Causality, which is a concept of marginal predictability.

- If a variable y_1 helps to forecast another variable y_2, given past y_2, then y_1 is said to *Ganger cause* y_2; otherwise, y_1 is said to *fail to (or does not) Granger cause* y_2 (Granger, 1969).

- y_1 and y_2 in the above definition can be generalized to groups of variables, $\mathbf{y}_1$ and $\mathbf{y}_2$.

- *Example:* Consider the bivariate VAR(p) model:

$$\begin{pmatrix} y_{1t} \\ y_{2t} \end{pmatrix} = \begin{pmatrix} \phi_{10} \\ \phi_{20} \end{pmatrix} + \begin{pmatrix} \phi_{11}^1 & \phi_{12}^1 \\ \phi_{21}^1 & \phi_{22}^1 \end{pmatrix} \begin{pmatrix} y_{1,t-1} \\ y_{2,t-1} \end{pmatrix} + \ldots + \begin{pmatrix} \phi_{11}^p & \phi_{12}^p \\ \phi_{21}^p & \phi_{22}^p \end{pmatrix} \begin{pmatrix} y_{1,t-p} \\ y_{2,t-p} \end{pmatrix} + \begin{pmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{pmatrix}.$$

y_2 fails to Granger cause y_1 iff $\phi_{12}^1 = \phi_{12}^2 = \ldots = \phi_{12}^p = 0$; y_1 fails to Granger cause y_2 iff $\phi_{21}^1 = \phi_{21}^2 = \ldots = \phi_{21}^p = 0$. Alternative, we can state the definition more succinctly using

the lag polynomial:

$$\begin{pmatrix} a(L) & b(L) \\ c(L) & d(L) \end{pmatrix} \begin{pmatrix} y_{1t} \\ y_{2t} \end{pmatrix} = \begin{pmatrix} \phi_{10} \\ \phi_{20} \end{pmatrix} + \begin{pmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{pmatrix},$$

where the expressions of the lag polynomials should be self-evident. Then y_2 fails to Granger cause y_1 iff $b(L) = 0$, that is, iff the autoregressive matrix lag polynomial is lower triangular.

- *Granger (non-)causality test:* For example, does y_2 Granger cause y_1? The null hypothesis is H_0: $\phi_{12}^1 = \phi_{12}^2 = \ldots = \phi_{12}^p = 0$. This can be tested using the standard F-test (assuming stationarity).

- Although sometimes the term "cause" was used as a shorthand for "Granger cause", it should be noted that the concept of Granger causality does not imply true causality in a deep sense; it only implies the *(linear) forecasting ability.*

- The underlying principles of Granger causality are: the cause precedes its effect, and it has unique information about the future values of its effects.

- Friedman and Kutter (1992) demonstrate the usefulness of Granger causality tests. The case for active monetary policy relies on a systematic link between the money supply and future price levels or real income. Estimating a VAR model with percentage changes in nominal income, money supply, and federal expenditures, they find that money supply Granger-caused nominal income before the late 1970s but not afterward.

2.7 Impulse Response Analysis

- Impulse response analysis helps quantify how a shock to one variable propagates through a system over time, revealing the dynamic relationships between variables in a VAR model.

- By tracing the effects of shocks, impulse response functions provide insights into the effectiveness of policy interventions and help assess the long-term impact of economic disturbances.